

Project Singularity: Towards generalized reasoning, integrating latent reasoning and world models

1. Introduction

1.1 Context and motivation

Our motivation for this project can be captured with the following question: Given our position as undergraduate students with foundational knowledge in machine learning, what meaningful progress can we make toward artificial general intelligence?

It is frequently asserted that Large Language Models (LLMs) don't really understand what they're being asked, that they don't truly reason, that they simply predict the next best tokens, based on training on massive data corpora, and that the reasoning and logic for token prediction is captured in a black box. Those are loose generalizations that do not fully capture the state of the art. The clarity of the mathematical operations underlying LLMs stands in stark contrast to our limited grasp of their emergent capabilities; a gap that serves not as a stopping point, but as a catalyst for a new generation of researchers to develop the mechanistic theories necessary to decode the 'black box' of artificial intelligence.

The first step, the "research phase" consisted of a focused literature review of modern AI / Machine Learning, learning about models, paradigms and methods. Some of the areas that retained our attention the most, and influenced our direction are Latent Space Reasoning, World Models, Hierarchical Reasoning Models, Universal Reasoning Models, Neurosymbolic Artificial Intelligence

1.2 But What is Reasoning?

The most central question to our objective. Answering this question is crucial because it helps us understand what it is we are trying to model. The American Psychological Association defines reasoning *as thinking in which logical processes of an inductive or deductive character are used to draw conclusions from facts or premises*. Thinking Fast and Slow, Daniel Kahneman's famous book on cognitive processes, represents the human reasoning machine by dividing it into the "Fast system", source of intuition, associated with low effort, involuntary thought processes and the "Slow system", temple of formal reasoning and intense cognitive activity, corresponding to effortful, conscient thinking, novel and complex task processing. Mentalese is a hypothetical, innate, and non-linguistic system of representation in the brain, often called the "language of thought". Behavioral and Neuroscience also provide helpful frameworks and insights.

There exist many representations of the human mind and of cognitive processes. A baseline understanding of those is very helpful to understand gaps, and unlock creative, meaning full approaches to theoretical and applied problematics of AI, Machine Learning and reasoning. Humans have been the only (widely known and recognized) entity credited with the ability to think and reason deeply, properly speaking. Then, in the 1940's, computers were created, electronic devices conceived to execute calculations at a steady, fast pace. Complex abstractions were built on top of the binary operations running of those machines, and almost a hundred years later, those very machines and abstractions emerged as the first alternative candidate to the complex reasoning, the *Intelligence* that characterizes humans. Humans are also defined by their biological limitations and their implications; experts are rare and expensive; programs can be duplicated and instantiated massively and for cheap. Humans are unpredictable and change in many ways, programs behave deterministically by nature. Modern computers are orders of magnitudes ahead of the human brain in terms of information and computation processing. This contrast motivates a key hypothesis:

If we can model the structure of human-like reasoning—without inheriting its biological limitations—we may achieve systems capable of scalable, general problem solving.

1.3 From Research to Design

Following our extensive literature review, the project transitioned to a solution phase. With everything that we have learned, how do we create a model capable of reasoning? One of the most distinctive aspects of human intelligence with respect to computers is its ability to discover and learn new tasks.

For this reason, we selected the competition ARC-AGI 3 as a primary benchmark. ARC-AGI 3 emphasizes abstraction, compositional reasoning, and requires generalization to unseen tasks, some of the main bottlenecks of AI, where the human brain is still far ahead of AI. Our participation to ARC-AGI is not in competitive, but rather to study the limitation of modern AI designs, including ours, and best performing concurrent models. Measuring, the rationale being that if our model(s) truly achieved generalizable reasoning, it should perform reasonably well on such tasks.

Two complementary directions emerged from this analysis. The first focuses on internal reasoning processes, where models iteratively refine latent representations through recursive computation. The second focuses on interaction with an environment,

where decision-making is framed as a sequence of actions evaluated through planning and search. These directions correspond, respectively, to latent reasoning models and world model-based systems.

Based on these insights, the team selected two primary approaches for implementation: Universal or Hierarchical Reasoning Models, which operate through recursive transformations in latent space, and world model architectures that enable structured interaction with an environment through planning mechanisms

1.4 Scope and Contribution

This work does not claim to achieve AGI. Its contribution lies instead in structuring a path toward reasoning-centric AI systems. By combining insights from latent reasoning models and world model architectures, we propose designs that are both grounded in current research and implementable within the scope of this project.

The remainder of this paper presents the findings of the research phase, details the proposed system architectures, and outlines the next steps required to further develop these ideas.

2. Models architecture

2.1 Universal Reasoning Models (URMs): Singularus

The Universal Reasoning Model (URM) is used in this work as a latent reasoning framework rather than a fixed, task-specific architecture. Our implementation follows three defining principles: unified parameterization, recurrent inductive bias, and stable training through truncated backpropagation.

Unified Parameterization in our model enforces the use of a single shared set of weights across all reasoning steps and tasks. This design prevents the emergence of task-specific specialization and instead encourages the model to learn reusable transformations that generalize across different problem instances. Recurrent Inductive Bias is introduced by iteratively applying the same transformation to a latent state, allowing the model to progressively refine its internal representation rather than producing an output in a single pass. Truncated Backpropagation Through Loops (TBPTL) is used during training by restricting gradient flow to a subset of these iterations, improving both stability and computational efficiency while preserving the depth of the reasoning process.

Our architecture is structured around a nested loop mechanism operating over latent representations. The inner loop consists of a transformer-based reasoning block that is recursively applied multiple times to the same latent state. Each iteration refines the representation, enabling the model to construct increasingly structured intermediate states. Because the same parameters are reused at every step, the model trades parameter count for iterative computation, aligning with the objective of modeling reasoning as a process rather than a single transformation.

This inner reasoning process is embedded within an outer loop that enables higher-level refinement. The outer loop repeatedly invokes the inner loop, allowing the model to revisit and improve its reasoning trajectory. During training, an Adaptive Computation Time (ACT) mechanism is used to determine when sufficient reasoning has been achieved, enabling early termination of the outer loop and improving efficiency without constraining the model’s capacity to reason deeply.

To enhance representational capacity without significantly increasing computational cost, our implementation incorporates the ConvSwiGLU modification within each transformer block. After the MLP layer, a lightweight convolutional operation with a small kernel is applied to the latent representation, enabling localized interactions between neighboring features. This is particularly useful for structured inputs such as grid-based environments, where local relationships are critical for reasoning.

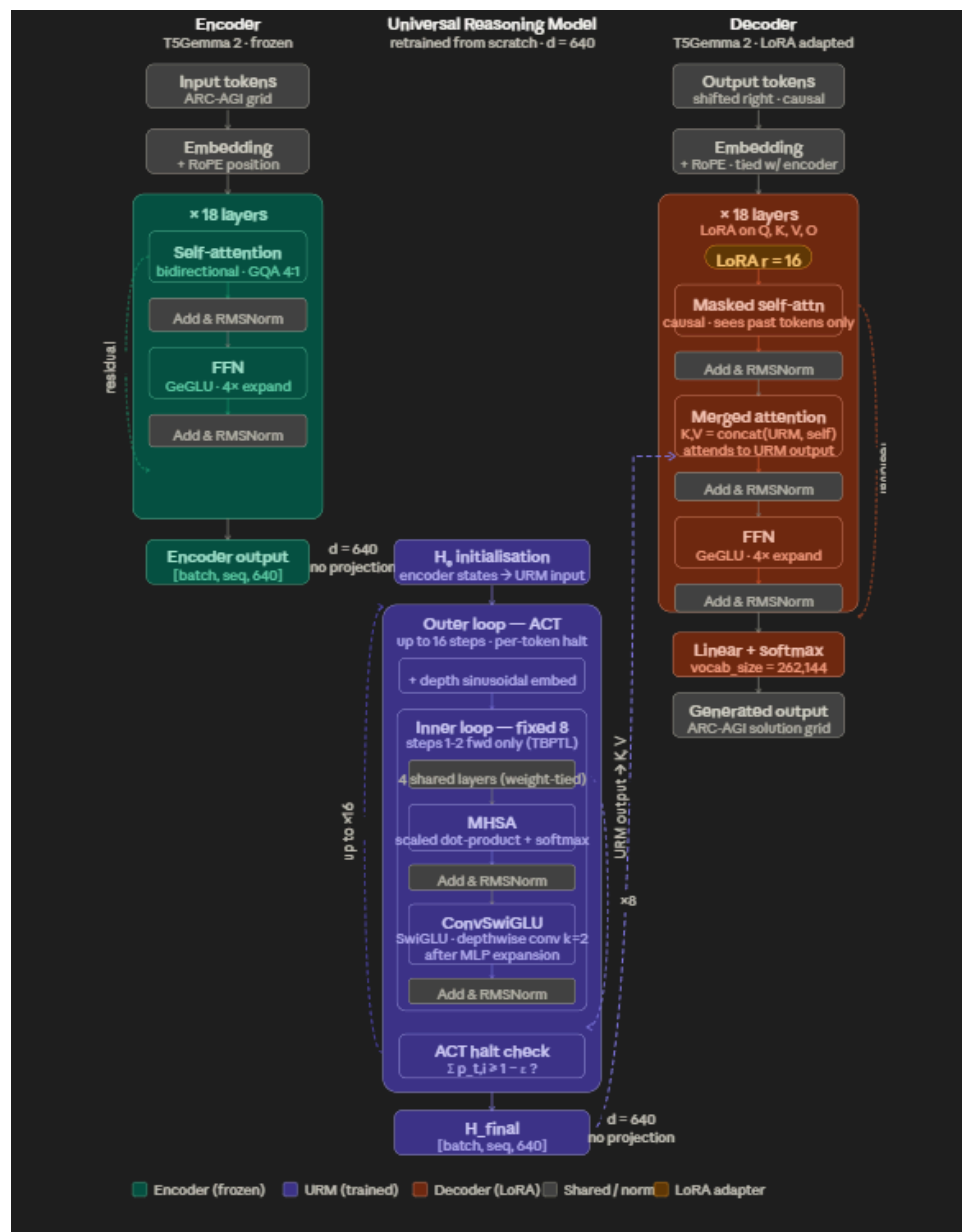
A key aspect of our design is the distinction between recursive computation and increased architectural depth. While deeper models introduce additional parameters, our approach enforces weight sharing across iterations, which constrains the model to reuse and refine its internal transformations. This constraint promotes abstraction and reduces reliance on memorization, which is essential for generalization in reasoning tasks.

For the ARC-style setting targeted in this project, the model is adapted from a direct input-output mapping to a sequential decision-making framework. Instead of producing a fully transformed output grid in a single pass, the model predicts the next action to apply to the current state. This allows the system to interact iteratively with the environment and aligns the architecture with step-based reasoning.

Additionally, we identify the absence of persistent memory as a limitation of the base URM framework and propose an extension through a latent memory mechanism. In this design, intermediate latent representations and action embeddings are stored and reused across reasoning steps, enabling the model to maintain context over longer horizons without relying on external symbolic representations.

Overall, our implementation of URM frames reasoning as an iterative, computation-driven process over shared representations. By combining weight sharing, recursive refinement, and controlled training dynamics, the model provides a compact yet expressive foundation for latent reasoning in structured environments.

Functional Schema of our Reasoning model,
Singularus



2.2 World model architecture

The world model architecture is built as a replay-based planning system for ARC in which action generation, evaluation, and search are separated into distinct modules. At the front of the pipeline, a proposer model receives the current ARC state and suggests a small set of plausible next actions instead of committing directly to a single move. In the current implementation, this proposer is powered by Gemini and operates not only on the raw frame and the currently valid actions, but also on a structured abstraction of the state, so that candidate generation is guided by both visual input and higher-level state descriptors. This design treats the world model less as a fully learned simulator and more as a model-guided action prior that narrows the search space before planning begins.

Once candidate actions are proposed, they are passed to a search procedure based on Monte Carlo Tree Search (MCTS). Rather than trusting the proposer in one shot, the system expands branches corresponding to several candidate actions and explores them using standard MCTS stages such as selection, expansion, simulation, and backpropagation. The search is guided by UCB-style exploration and can perform shallow rollouts so that the chosen move is informed not only by an immediate transition but also by a short prospective trajectory. At decision time, the architecture selects the root action with the strongest search-based value or visitation signal, making the final behavior a product of structured planning rather than direct generation alone.

A second key component is the critic, which evaluates how promising a transition appears after a candidate action is applied. Importantly, the critic does not score a state in isolation. Instead, it compares the previous frame and the resulting next frame, allowing it to reason about change, progress, and local improvement toward a heuristic objective. This transition-based evaluation gives the search algorithm a ranking signal for branches and helps compensate for the fact that the proposer itself is not a grounded dynamics model. In effect, the proposer generates plausible moves, the critic judges their consequences, and MCTS organizes which futures deserve deeper exploration.

The most important systems constraint in this architecture is that the environment cannot be cheaply branched. The agent cannot simply clone the current environment state at every node and simulate children independently. Because of this, each hypothetical branch must be reconstructed by resetting the environment to the root and replaying the full sequence of actions that leads to the node under consideration. This constraint strongly shaped the implementation: each MCTS node stores its full action sequence, and replaying that sequence is central to every deeper evaluation. As a

result, the architecture is not just model-guided MCTS, but specifically replay-based model-guided MCTS.

To make this replay-heavy planning practical, the project also introduced an environment refactor. The ARC interface was redesigned into a backend-agnostic wrapper that can operate either through the live HTTP API or through a local offline toolkit. This change allowed exactly the same planning logic to run in both settings and significantly accelerated iteration, since frequent reset-and-replay becomes far more manageable in a local environment than against an online service alone. That engineering layer is therefore part of the architecture in a real sense: without it, the search procedure would be too slow and cumbersome to develop effectively.

Another architectural improvement is the addition of a state abstraction layer. Rather than feeding only a raw 64×64 grid to the proposer and critic, the system extracts structured features such as connected components, value counts, bounding boxes, centroids, border contact, and transition-level deltas like changed cells and changed value pairs. These abstractions provide a more compact and interpretable description of both state and state change, helping the LLM components reason over the environment in a more structured way. This moves the pipeline beyond a bare prompt-over-pixels setup and toward a hybrid architecture in which symbolic descriptors support search-based decision making.

Overall, the architecture can be understood as a modular planning stack for ARC: a world-model proxy proposes actions, a critic evaluates transitions, and replay-based MCTS performs multi-step planning under strict environment constraints. The broader research document presents this as a minimal but feasible proof of concept, while the implementation notes in the message file make clear that the real contribution lies in turning those abstract pieces into a functioning system under reset-and-replay limitations. In that sense, the architecture is less about claiming a fully learned world model and more about demonstrating a workable path toward model-guided reasoning in ARC-style environments.

3. Next Steps

The current work establishes two complementary directions: a latent reasoning framework based on the Universal Reasoning Model (URM), and a planning-based pipeline approximating a World Model through search and external evaluators. Both demonstrate feasibility at a conceptual and prototypical level, but neither yet achieves the level of abstraction, grounding, or learning required for robust performance on ARC-AGI-3. The same can be said of the other competing models.

For the URM track, the next step is to move beyond task-bound transformation and toward domain-agnostic reasoning. While the recursive structure provides a strong inductive bias for iterative computation, its current formulation remains tied to fixed input-output mappings. To improve generality, the model must instead operate as a sequential decision process over latent states, where reasoning unfolds through intermediate updates rather than a single transformation. This shift enables the model to construct solutions incrementally and adapt its computation based on evolving internal representations. Crucially, this must be supported by persistent latent memory, allowing the system to retain and manipulate hypotheses across steps. Such a mechanism would elevate the recursion from repeated computation to structured reasoning, enabling the reuse of learned inference patterns across diverse tasks.

For the World Model pipeline, the central improvement lies in internalizing and generalizing its understanding of environment dynamics. The current reliance on external language models for action generation and evaluation limits both consistency and scalability. A more general formulation would learn compact, compositional representations of states and transitions, enabling the system to simulate and evaluate possible futures within its own latent space. This would allow planning to emerge from an internal predictive model rather than external heuristics, supporting adaptation to new environments, rules, or objectives. The emphasis shifts from solving a predefined task space to acquiring reusable predictive structure, where knowledge about actions and consequences can transfer across domains.

A shared limitation across both approaches is the lack of structured abstraction. Operating directly on raw grids constrains both reasoning and planning, as it prevents the emergence of stable object-level or relational representations. Addressing this requires the introduction of a structured latent space that captures entities, transformations, and relationships in a compositional manner. Such representations would serve as a common interface between reasoning and planning, allowing both URM and World Model components to operate over higher-level abstractions rather than low-level encodings.

Neurosymbolic integration provides a natural pathway to achieve this level of structure and generalization. By combining neural representation learning with symbolic constraints and abstractions, the system can leverage both flexibility and compositional structure. Neural components enable generalization from data and adaptation to new

inputs, while symbolic elements introduce explicit representations of objects, relations, and rules that remain stable across tasks. In the URM context, this can guide the organization and manipulation of latent memory, enforcing consistency in multi-step reasoning. In the World Model context, it can constrain action spaces, encode valid transformations, and structure planning over interpretable entities. More broadly, this integration enables the system to move beyond pattern recognition toward systematic reasoning, which is essential for general intelligence.

Finally, integration remains the central objective. Rather than treating the different models and prototypes implemented as independent directions, the long term goal is to unify them into a coherent architecture. Together, they would form a system capable of constructing, testing, and refining solutions in a loop, combining internal reasoning with external grounding.